DS 208 · GRADUATE · UP CEBU

Welcome to Programming for Data Science

Code that runs once and gives you a number is not worth much. We are after code someone else can run next year and get the same number.

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12 weeks	Saturdays	Fully online	Start at zero
BASICS TO PROJECT	THREE HOURS	SLIDES STAY UP	NO EXPERIENCE NEEDED

The whole plan is on the next slide. Nothing on it is a surprise later.

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THE PLAN

Twelve weeks, four stretches

The first month is the steepest. After that you are mostly learning libraries, which is much easier than learning to program.

WEEKS 1–4	**The language itself**	Setting up Python, then types, control flow, functions, data structures and files. Week 4 is where code stops being one long script — objects, errors, and how to organise a project.
WEEKS 5–8	**The libraries you will actually use**	NumPy for arrays, then two weeks of pandas because it earns them — grouping, merging, reshaping. Then matplotlib and seaborn, so the result is something you can show a person.
WEEKS 9–11	**Data that lives somewhere else**	Reading from files, SQL and APIs. Regular expressions for text that refuses to be tidy. Then Git, environments and debugging — the habits that make Week 12 survivable.
WEEK 12	**Your turn**	One project that uses most of the above, and you present it.

WHAT THIS COURSE IS

Programming as analysis, not computer science

You are not here to become a software engineer. You are here because a spreadsheet eventually stops being enough, and code is what comes next.

Reproducible
A script states exactly what you did. Next month, it still says the same thing.

Scalable
The work is the same whether the file has a hundred rows or a million.

Checkable
Someone else can read your analysis and find your mistake — which is the point.

WHAT YOU LEAVE WITH

What you should be able to do by Week 12

Not “know Python”. That phrase means nothing. These four are checkable.

Read someone else’s code

Open an unfamiliar script and work out what it does. You will do this far more often than you write anything from scratch.

Move data without leaving the keyboard

Load, filter, group, join and reshape a real dataset in pandas, instead of clicking through a spreadsheet and hoping.

Make a figure that argues something

Not a default chart with a default title. One where the choice of chart is itself the point being made.

Hand your work to someone else

A project with pinned dependencies and a README that runs on a machine which is not yours.

HOW WE WORK

Fully online — and that changes the rules

We meet on Saturdays for three contact hours. Everything around that is yours to schedule.

Everything lives on this portal

Slides for every week stay here permanently. Nothing important is only said out loud, so missing a session is recoverable.

Get stuck for 30 minutes, then ask

Struggling is how this is learned — but struggling alone for three days is not. There is a limit, and it is half an hour.

WHO IS IN THE ROOM

Zero programming experience is fine

There are no prerequisites beyond graduate standing. We begin at types and control flow in Week 2 — genuinely from the start.

If you have never coded

You are the student this course is designed around. Nothing assumes you have seen a terminal before.

If you already code

The novelty is the data stack — NumPy, pandas, matplotlib — and the habits that make analysis reproducible.

DOING WELL HERE

Four habits that decide how this goes

Programming is learned by hand. There is no version of this where you watch and absorb it.

Type it, do not read it

Retype the examples — including the parts you are sure you already understand. That is where the surprises are.

Break things on purpose

Change a line, predict what will happen, then run it. Being wrong about the prediction is the whole lesson.

Read the error

The traceback names the file and the line. Most people scroll past the one piece of information they needed.

Start Week 12 in Week 8

The project is not a week of work. The people who begin early are the ones who finish calm.

BEFORE OUR FIRST SESSION

Get Python running — the easy way first

Start in the browser. Nothing to install, nothing to break, and it works the same on every machine.

1 · Open Colab

Go to **colab.research.google.com** and sign in with a Google account.

2 · Run one cell

New notebook, type **print("hello")**, press Shift+Enter. That is the whole task.

3 · Optional: Anaconda

Want a local setup? Install Anaconda too — but Colab alone is enough to start.

NEXT

Week 1 · Python & the Data Science Toolkit

We start where the spreadsheet gives up, sort out which library does what, and spend real time on error messages. You will read a lot of those this semester, so you may as well stop fearing them in week one.

Week 1 slides are already on the course page. Read ahead if you want to.